

Programme	B. Sc. Mathematics Honours			
Course Code	MAT8CJ489			
Course Title	RESEARCH METHODOLOGY IN MATHEMATICS			
Type of Course	Major			
Semester	VII			
Academic Level	400 – 499			
Course Details	Credit	Lecture/Tutorial per week	Practicum per week	Total Hours
	4	4	-	60
Pre-requisites	1. Mathematical Logic and necessary exposure to set theory. 2. Research Aptitude			
Course Summary	MAT8CJ489, "Research Methodology in Mathematics," is designed to equip students with the essential skills and knowledge required for conducting research in mathematics effectively. This course focuses on various aspects of mathematical research, including axiomatic set theory, writing mathematics, researching and presenting findings, and using LaTeX for mathematical typesetting. Additionally, students explore open-ended research topics, allowing them to delve into specific areas of interest within mathematics. Throughout the course, students engage with key texts and resources, enabling them to develop a comprehensive understanding of research methodologies in mathematics.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Set Theory and Mathematical Writing: Students will demonstrate proficiency in axiomatic set theory, including concepts such as relations, functions, and Peano axioms. Students will exhibit competence in mathematical writing.	Ap	C	Internal Examination/ Assignment/ End Sem examination
CO2	Research Skills and Presentation Techniques: Students will acquire research skills, including identifying research topics. Students will develop effective presentation techniques, giving talks.	Ap	P	Internal examination/ Seminar/ Assignment/ End Sem examination
CO3	Mathematical typesetting: to use LaTeX to create and typeset documents. Beamer Presentations and PSTricks also included.	Ap	P	Internal Examination/Seminar/ Assignment/End Sem examination
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text Book	(1): Naive set theory: Paul R. Halmos, Courier Dover Publications, 2017. (2): A student's guide to the study, practice, and tools of modern mathematics, Donald Bindner and Martin Erickson. CRC Press, ISBN: 978-1-4398-4606-3			
Module	Unit	Content	Hrs (48+12)	External Marks (70)
I	Axiomatic Set Theory		12	
		(Sections 1 to 12 from the Text 1.) 1: The axiom of extension 2: The axiom of specification 3: Unordered pairs 4: Unions and intersections 5: Complements and powers 6: Ordered pairs 7: Relations 8: Functions 9: Families 10: Inverses and composites 11: Numbers 12: The Peano axioms		
II	Writing Mathematics (Text 2)		12	
		Chapter 1: How to Learn Mathematics (A quick review – not part of evaluation)		
		Chapter 2: How to Write Mathematics - 2.1: What is the goal of mathematical writing? 2.2: General principles of mathematical writing 2.3: Writing mathematical sentences 2.4: Avoiding error 2.5: Writing mathematical solutions and proofs		

		2.6: Writing longer mathematical works 2.7: The revision process		
III	Researching and Presenting (Text 2)		12	
		Chapter 3: How to Research Mathematics - 3.1: What is mathematical research? 3.2: Finding a research topic 3.3: General advice 3.4: Taking basic steps 3.5: Fixing common problems 3.6: Using computer resources 3.7: Practicing good mathematical judgment Chapter 4: How to Present Mathematics - 4.1: Why give a presentation of mathematics? 4.2: Preparing your talk 4.3: DOs and DON'Ts 4.4: Using technology 4.5: Answering questions 4.6: Publishing your research		
IV	LATEX (Text 2)		12	
		LaTeX 9.4 How to create and typeset a simple LATEX document 9.5 How to add basic information to your document 9.6 How to do elementary mathematical typesetting 9.7 How to do advanced mathematical typesetting 9.8 How to use graphics		
		PsTricks		

	10.1 What is PSTricks? 10.2 How to make simple pictures 10.3 How to plot functions 10.4 How to make pictures with nodes		
	Beamer 11.1 What is Beamer? 11.2 How to think in terms of frames 11.3 How to set up a Beamer document 11.4 How to enhance a Beamer presentation		
V	OPEN ENDED (General Mathematical Research) Lecturer's choices from the following Reference 1 (Princeton Companion), Section 1.4: General Goals of Mathematical Research, p.48 to 78. 1. Solving Equations 2. Classifying 3. Generalizing 4. Discovering Patterns 5. Explaining Apparent Coincidences 6. Counting and Measuring 7. Determining Whether Different Mathematical Properties are Compatible 8. Working with Arguments that are not Fully Rigorous 9. Finding Explicit Proofs and Algorithms 10. What do you find in a Mathematical Paper? Reference 2 (Math Unlimited), any chapters of the lecturer's choices. Reference 3 (Krantz, Mathematical Writing), any topics of lecturer's choice.	12	
Reference	1. The Princeton companion to mathematics, Timothy Gowers, Ed., Princeton University Press, 2008, ISBN ISBN 978-0-691-11880-2. 2. Math Unlimited, Essays in Mathematics, Editors: R. Sujatha, H N Ramaswamy, C S Yogananda, CRC Press, 2012, ISBN: 978-1-57808-704-4. 3. A Primer of Mathematical Writing, Steven G. Krantz, 2nd Ed., 2017, ISBN 9781470436582.		

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	1	1	2	3	2	3	2	3	1	2
CO 2	1	2	0	3	3	3	3	2	3	1	3
CO 3	0	1	3	1	2	2	3	3	2	1	2

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓	✓	✓
CO 3	✓	✓	✓	✓

MULTI-DISCIPLINARY COURSES
(MDC)